

Jamming Attacks in 5G NR FR1

Péter János Varga
Óbuda University
Kandó Kálmán Faculty of
Electrical Engineering
Budapest, Hungary
orcid.org/0000-0002-4964-7249

Tibor Wüthrl
Óbuda University
Kandó Kálmán Faculty of
Electrical Engineering
Budapest, Hungary
orcid.org/0000-0002-7522-3511

Sándor Gyányi
Óbuda University
Kandó Kálmán Faculty of
Electrical Engineering
Budapest, Hungary
orcid.org/0000-0001-9405-3864

Márk Tamás Baross
Óbuda University
Kandó Kálmán Faculty of
Electrical Engineering
Budapest, Hungary
orcid.org/0000-0002-8330-0302

Attila Németh
Óbuda University
Doctoral School on Safety and
Security Sciences
Budapest, Hungary
orcid.org/0000-0002-0688-7381

Abstract—Mobile communication has become an integral part of our everyday life, nowadays the fifth generation (5G) solutions represent the high end of technology. In Hungary and in many other countries around the world, phone carriers and service providers have started to build their own fifth generation networks alongside the second (2G) and fourth generation (4G) solutions. In this article, we examine how a current fifth-generation network and its radio segment work against of electronic disturbance. Our team conducted experiments at the Infocommunication and Information Technologies National Laboratory, within the Ministry of Innovation and Technology, National Research, Development and Innovation Office.

Keywords— Mobile communication; 5G; NR; FR1; jamming,

I. INTRODUCTION

The experiments in the research described in this article focused on the radio disturbance in 5G networks. We created a test environment, where we tried to validate our assumptions by conducting tests and measurements. To reach this goal, our team chose the necessary technology and built the experimental environment. Reveal a radio interface vulnerability was the primary goal of these tests. A serious vulnerability can ultimately degrade a well-functioning 5G network completely inoperable.[1]

Our research focused on 5G Stand Alone (SA) networks, although we believe that the results of our experiments will be useful even in the case of Non-standalone (NSA) deployment. By conducting experiments, we were examined the 5G network behavior against different jamming techniques.[2,3]

II. 5G NEW RADIO (NR) RELEVANT DETAILS

A. 5G SA NR structure and frequency range

The 2019 "Release 15" version published by 3GPP defines the regulation of radio parameters of 5G networks. The parameters of the 5G SA network examined by our team in the experiments were also specified in this standard. The Figure 1. shows the architecture of a typical 5G SA. [4,5]



Fig. 1. 5G SA structure

The standard defines two distinct frequency ranges for 5G:

- FR1: 450 MHz – 7125 MHz and
- FR2: 24250 MHz - 52600 MHz.

The tests were performed in the FR1 band. In this range, usually “macrocells” are used to cover a larger area, while covering smaller areas are possible by using “picocell”. Figure 2. shows the pre-set parameters of the test system in the FR1 band. [6]

TABLE I. 5G FR1 NETWORK PARAMETERS

Frequency range	Frequency band	Descriptive frequency	UL and DL frequency range	Duplex
FR1 (3–6 GHz)	n78	3.6 GHz	3300–3800	TDD

Fig. 2. 5G FR1 parameters

B. Network configuration for jamming

To implementing the target and the jamming system, the following hardware devices were used:

- 5G SA network;
- USRP B210 SDR;
- 5G routers;
- PCs for client and server roles;
- 5G Mobile phone;
- Spectrum analyzer.

USRP B210 SDR were used for jamming, this type of software radio can be used between 70MHz and 6GHz in full duplex mode, with a bandwidth of 56 MHz. [7] It has open source drivers, the SDR device was controlled in a Windows environment using the GNU Radio application. Three different devices were connected to the 5G SA network: a 5G-capable smartphone and two personal computers. One of these PC acted as the client and the other were used as a server, both were connected through 5G-capable routers. Ubuntu Linux operating system were used on these computers. For measuring the maximum bandwidth of the IP networks were measured by using the iPerf3 network test application on smartphones and computers. The Figure 3. shows the structure of the jamming system. [8,9]

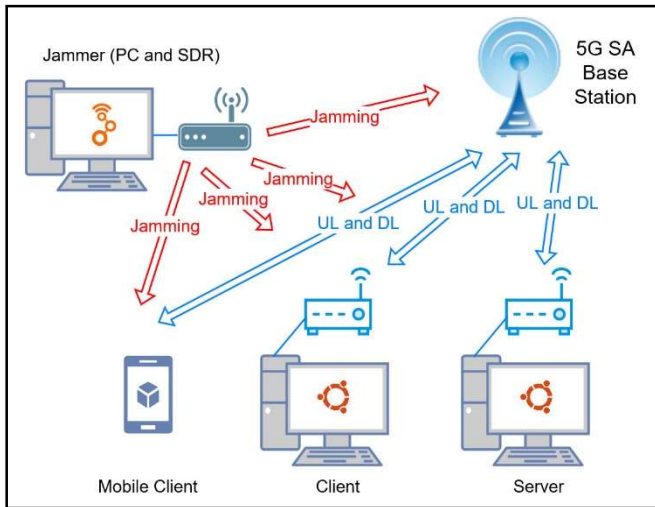


Fig. 3. Network configuration for jamming

III. RADIO JAMMING ATTACKS

The main goal of radio jamming in 5G networks is to prevent communication between the 5G cell subsystems and the devices connected to the network. One of the possible way to doing this using traditional radio jamming techniques in the physical layer. The aim of this technique is always to degrade the signal-to-noise ratio to reach a certain limit where the information reception in the target devices – basically, the demodulation of the received radio signal – becomes impossible, or at least problematic. Radio components of the cell itself or devices connected to cell can be targets of jamming. During a possible attacking event on 5G base stations, the disrupted network segment (cell) becomes inoperable, thus all devices in that cell cannot connect to the system. Another possible methods could target the client devices – mobile terminals, smartphones, etc. – that want to connect to the network. Attacking these devices, the attacker must select specialized methods for each kind of targets, probably these attacks do not cause full degradation of the whole network segment (cell). [10,11]

Of course, the effectiveness of radio jamming always depends on the power ratio of the jamming signal (which represents the “noise”) and the “normal” signal. Our experiments were performed in a 5G SA cell where the signal strength of the “normal” signal was around -26 dBm, while the maximum signal strength of the jamming signal was -50 dBm. During the test, the “client” devices were 5 meters away from each other and from the 5G antenna of the cell. Different jamming methods were used for the experiment. For each jamming technique, the needed parameters were the center frequency of the 5G radio channel, the channel bandwidth and the signal strength of the transmitter. During the simulated attack, the parameters of the jamming source must be adjusted according to these variables. Multiple jamming methods were used. [12,13]

A. Barrage jamming

This technique is a high-power, continuous, high-bandwidth jamming. Its aim is to prevent access to the radio channel. [14]

B. Spot jamming

During spot jamming, the jamming equipment can transmit signal with high power density in a narrow band. This method could completely disrupt a part of the band. [14]

C. Sweep jamming

In this jamming method, the jamming signal sweeps the entire bandwidth. To achieve this, the jamming device continuously tunes the transmitter frequency between specified values. [14]

IV. JAMMING RESULTS

The tests were performed using the two computers. Before the jamming, the network operational state and parameters were examined using the spectrum analyzer. During the tests, the radio cell had the following radio parameters:

- center frequency: 3.428120 GHz;
- bandwidth: 40 MHz;
- maximum signal level: -26 dBm

A. Reference performance

The reference measurement was performed in two steps. Firstly, we examined the spectrum of the disturbance-free signal, Figure 4. shows the reference spectrum.

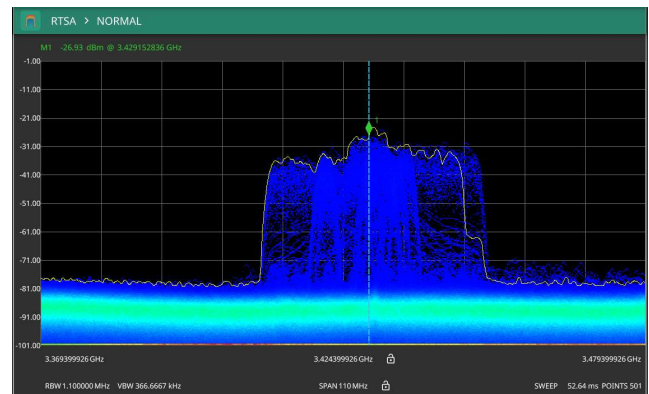


Fig. 4. Reference performance spectrum

The spectrum on the image clearly indicates the correct, preset parameters of our radio signal.

In the second step, the data transmission parameters of the network were tested and measured. To perform this examination, we used two different setups. In the first setup, the client computer and the server communicated with each other via 5G routers, while in the second case the mobile phone played as the client and communicated with the server on the 5G network. In both cases, the iPerf3 application was the data recorder, we captured the results of the 5 measurements. Then these raw data were processed and displayed using a graph. On the graph, the x axis is time, y axis is download or upload data transfer speed. The Figure 5. shows the measurement results of the communication between client and server computers. [15]

In the chart, the blue line indicates the download speed, while the orange line shows the upload speed in Mbit/s unit. This figure clearly shows that the download and upload speed tendencies were consistent, with little deviation. This test were repeated by using the mobile device as the client, data captured with similar method.

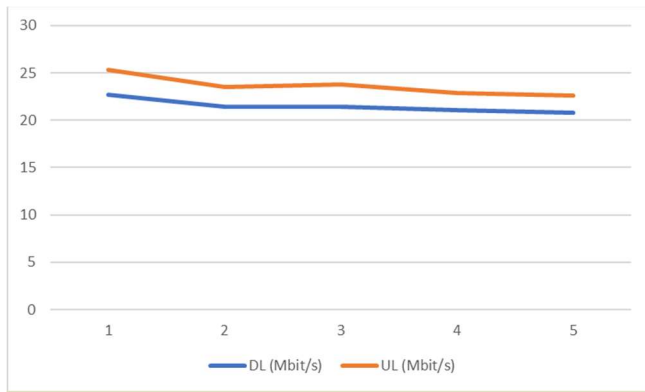


Fig. 5. Reference performance – Client to Server communication

Figure 6. shows the graph based on the processed results..

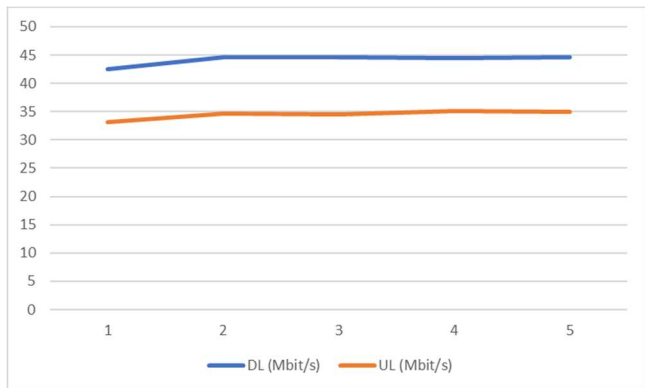


Fig. 6. Reference performance – Mobile to Server communication

As in the previous step, the upload and download speed tendencies are similar, with one difference: here we measured higher download speed values while in the first case, upload speed was higher.

B. Spot jamming

- Lower part of the operational band

When creating the jamming system, the GNU Radio application as the software environment was chosen. To build the spot jammer, we had to select the required blocks and set the software parameters up. Figure 7. shows the software structure of the spot jammer.

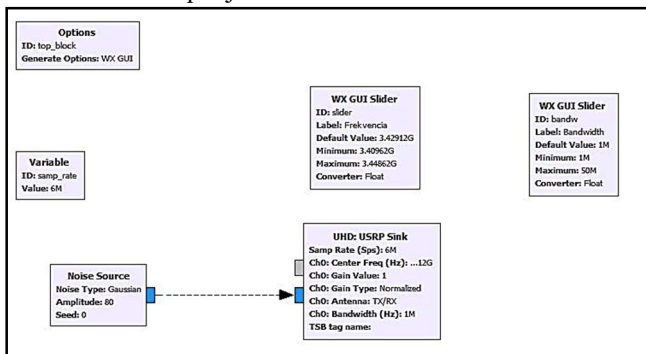


Fig. 7. GNU Radio – Spot structure

In the first jamming test, we emitted a jamming signal with a bandwidth of 1 MHz. The base frequency of the jamming signal was 3.22912 GHz – at the lower end of the channel. The spectrum image of this test measured by the spectrum analyzer is shown in Figure 8. [15]

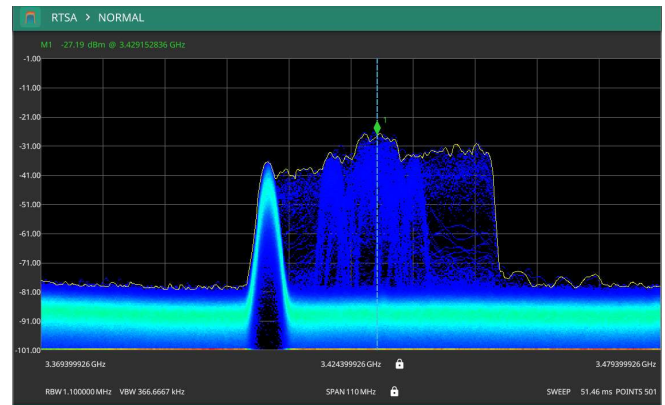


Fig. 8. Spot jamming – Lower part of the operational band

In this spectrum, the jamming signal spectrum at the beginning of the band is clearly visible, next to the “normal” signal. In this jamming setup, the network parameters between the two computers could be examined, because the mobile phone lost the network connection. The chart, containing the download and upload speed result of this test is shown in Figure 9.

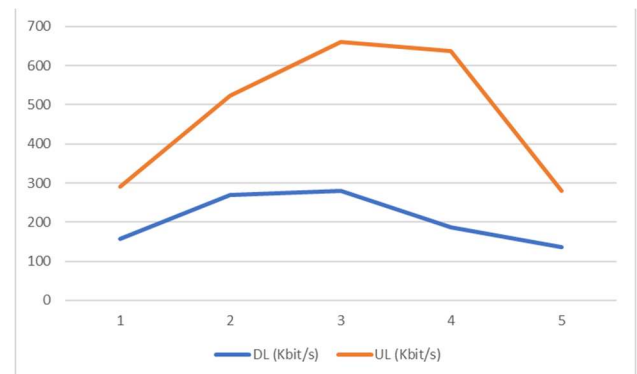


Fig. 9. Performance – Client to Server communication (Spot jamming – Lower part of the operational band)

As the values in the chart clearly shows, the download and upload ratio and tendency has changed, compared to “noise-free”, undisturbed environment. The diagram uses kbit/s units for both download and upload speeds.

- Middle of the band

Spot jamming test was repeated, but in the middle of the band - at 3.42912 GHz - with a 1 MHz jamming signal. The Figure 10. shows the spectrum image during operation of the jammer.

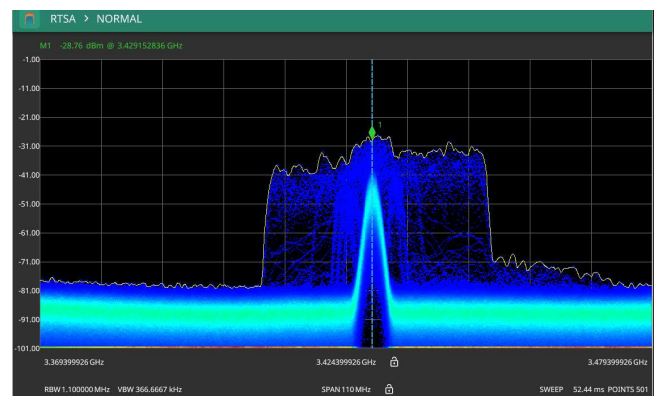


Fig. 10. Spot jamming – middle of the band

The jammer signal is clearly visible in the middle of the spectrum diagram of the cell operation channel band. The test of the network parameters for this set-up could not be performed, because the devices lost their mobile network connection.

- Higher part of the operational band

In this test the jamming signal was located at the end of the band - at 3.62912 GHz. The bandwidth of the disturbance signal was kept at 1 MHz. Figure 11. shows the spectrum image of this experiment.

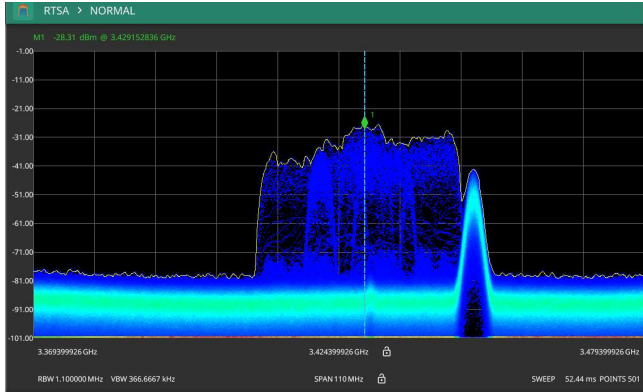


Fig. 11. Spot jamming – Higher part of the band

In this spectrum image, the spectrum of the jammer signal is clearly visible at the high end of the band. Using this jamming setup, we were able to measure the network parameters for both configurations – between computer client and the server and between mobile phone and server. Figure 12. shows the tendency of the download and upload data speed values of the communication between client and server computers.

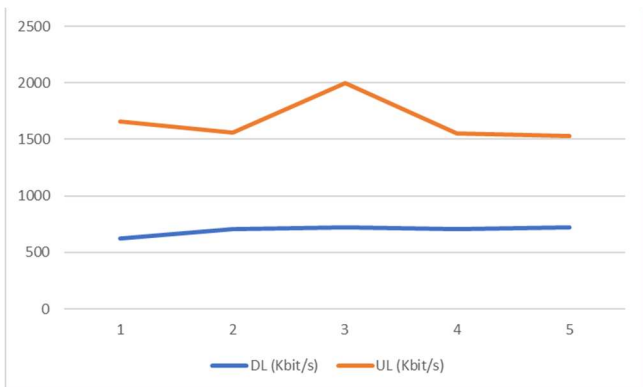


Fig. 12. Performance – Client to Server communication (Spot jamming – Higher part of the band)

It is clear that the download and upload ratio has changed compared to “noise-free”, undisturbed values. The speed values degraded, in the diagram Kbit/s units were used for both download and upload speeds. Figure 13. shows the tendency of the communication speed between the mobile device and the server.

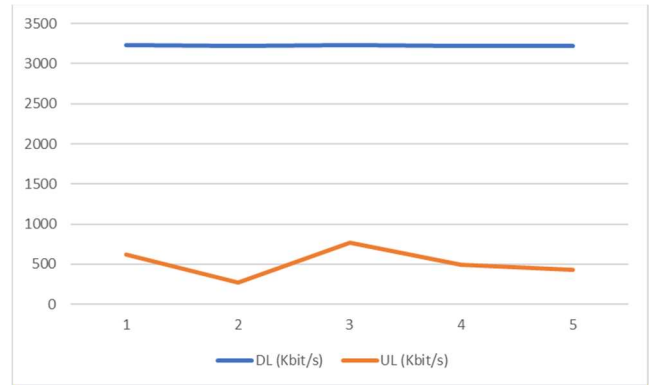


Fig. 13. Performance – Mobile to Server communication (Spot jamming – Higher part of the band)

As the diagram clearly shows that the upload and download speeds were also degraded here, values are in the order of Kbit/s. The ratio between upload and download speeds were about the same in the “noise-free” and disturbed environment, however, this ratio is different compared to the case of communication between the client and server. In case of client-server computer communication the upload speed is higher, while in the case of the connection between the mobile device and the server, the download speed is higher.

C. Barrage jamming

In the barrage jammer configuration the settings were different in two parameters compared to spot jammer setups. One of these parameter was the center frequency. In the barrage jamming test, the center frequency was 3.42912 GHz while the spot jammer setups used different frequencies on different part of the channel band. The other significant change was the bandwidth of our noise source, which in the barrage jammer setup was 40 MHz compared to the 1MHz in the spot jamming versions. The SDR device is capable of emitting a signal with larger bandwidth, so during the test 50MHz bandwidth had been used, to cover the whole band. Figure 14. shows the software structure of the barrage jammer setup. [16]

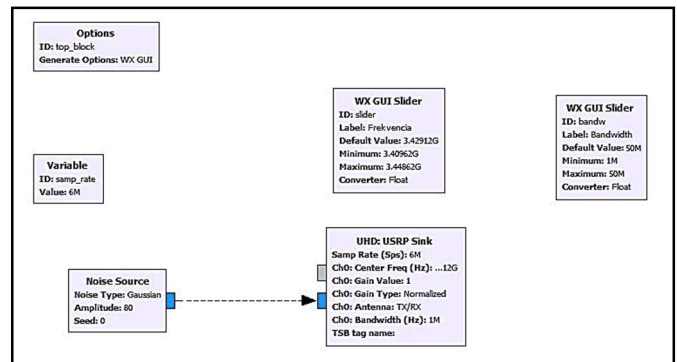


Fig. 14. GNU Radio – Barrage structure

In this type of jamming, the jamming signal suppresses completely the “normal” signal in most cases. Figure 15. illustrates the spectrum image of the disturbed channel.

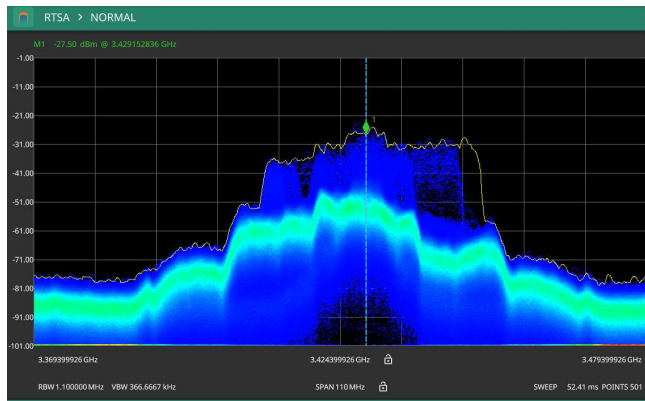


Fig. 15. Barrage jamming

The diagram clearly illustrates how the jamming signal affects the entire 5G spectrum of the target channel. The spectral image of our disturbance is clearly visible, the whole band has been affected by the noise. As a result of this test, mobile phones and 5G routers used by the client and server computers also lost their network connection completely so no measurements of communication speed could be performed.

D. Sweep jamming

We had to use several components in GNU Radio for our sweep jammer setup. As in the previous cases, we had to set the basic parameters. For this jamming mode, these parameters were the following:

- start frequency: 3.22912 GHz;
- stop frequency: 3.62912 GHz;
- jamming signal bandwidth: 1 MHz;
- sweep time: 1ms, 10ms, 30ms.

Figure 16. shows the GNU Radio software structure of the sweep jammer.

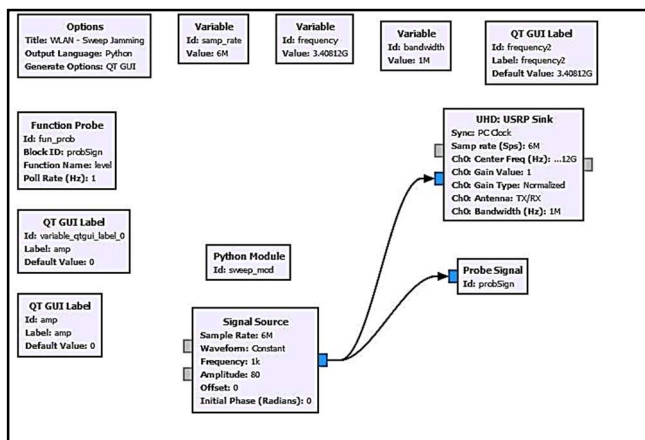


Fig. 16. GNU Radio – Sweep structure

Figure 17. shows the radio spectrum during the sweep jammer test.

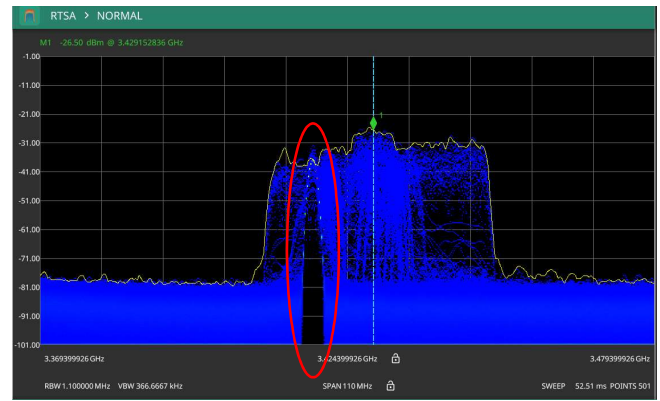


Fig. 17. Sweep jamming

Because the sweep jammer dynamically changed the jamming signal frequency during the operation, the spectrum image is just a snapshot. This measurement was repeated using different sweep times: 1 ms, 10 ms and 30 ms. As a result, in all three sweep times used, mobile phones and 5G routers lost their network connection completely.

CONCLUSIONS

Mobile networks and the connected devices are exposed to electronic disturbance all the time, intentionally or unintentionally. In the case of intentional disturbance, the user of the 5G service does not perceive the jamming directly, only the results of it. During the intentional jamming, service quality could be degraded, even the entire radio channel might become unusable for users in the mobile network. In our article, we demonstrated some jamming methods and their results in a 5G network. As a conclusion of these experiments, we were able to prove that with the help of a simple, commercially available Software Defined Radio hardware with an open source software component, these disturbances in the physical layer can be performed with high efficiency by almost anyone. Of course, the jamming methods presented in our case studies can be detected and measured by using the right equipment. On the other hand, more precise and shorter jamming can be implemented with the help of SDR software, in which case the attacker could make the 5G NR base station interface inaccessible only for data streams critical to operation. The detection of such disturbances could have significant difficulties, meaning that the disturbance may remain operational for a long time.

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